

Unit 107 Field Data Review

Mohawk Valley West Slope demonstration. **Tractor** unit, 95.3 ac. 46 plots, BAF 20, variable-radius cruise on a 300 ft grid, 522 trees tallied. Simulated data with planted recording errors; see docs/methods.md. Plots tally all trees, so this is a pre-marking stand exam; the sale-cruise standards are applied as if the whole tallied volume were designated for cutting.

TABLE 1. STAND SUMMARY, BAF 20 CRUISE, PER ACRE

STAND DENSITY		STAND INFO	
Basal area	227 sq ft/ac (1 SE of mean, BA 1.6 %)	Type	Sierran mixed conifer
Trees per acre	72	CWHR size / density	4 / D (cover 93 %)
QMD	24.0 in	SDI	280 (48 % of max 584)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	226 sq ft/ac 24.4 in	BA QMD	1 sq ft/ac 8.9 in
TPA	69	TPA	3
Volume	39.6 MBF/ac (7889 cu ft)	Volume	0.6 green tons/ac (25 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI, about 165 sq ft/ac BA	Vol. sampling error	6.8 % at 95 % (CV 23 %, t = 2.01, n = 46)
Species (BA)	WF 71, PP 65, IC 37, DF 33, SP 21	Stratum standard	30 % of volume (R5 2409.12-2021-4, sec. 41.1): MEETS ; plots for standard 3
		Sale volume share	6.9 % of sale volume: standard shown for information (under the R5 10 % threshold)
		BA sampling error	3.3 % at 95 % (CV 11 %); reference, not the standard
		Measured trees	PP 149, WF 164, DF 76, SP 48, IC 85; R5 20 measured trees per major species (sec. 41.3; major by sale value: PP, WF, DF, IC): ok
		Plot count	46; R5 20-plot minimum for an area-based cruise (sec. 41.3): ok

Cubic volume is total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium), net of defect. Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006). Green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber; applied to total-stem CVTS it overstates sawlog volume. Merchantable cubic volume is the consistent pairing. Sampling error at 95 % confidence, Student's t, on plot net cubic volume per acre, the quantity the standards test. Standards: Region 5 Supplement 2409.12-2021-4 to FSH 2409.12 ch. 40 (effective 2021-08-11), sec. 41.1, stratum standard 30 % of volume for tree-measurement sales (40 % scaled; the national text 41.1(5)(b) reads 40 %), binding for any stratum over 10 % of total sale volume and shown for information otherwise; sec. 41.3, 20 measured trees per major species (10 % or more of sale value; one stumpage rate, so value share equals volume share) and 20 plots minimum. Measured trees are tally trees with a valid code, DBH in range and a height. The basal-area error is reference only. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables drop records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

TABLE 2. STAND TABLE BY 4-INCH DBH CLASS, PER ACRE

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	7	5.6	3.0	63	0.3
12-15.9	33	13.0	14.3	327	1.6
16-19.9	62	15.5	27.0	722	3.6
20-23.9	79	13.5	34.3	1049	5.3
24-27.9	82	9.9	35.7	1203	6.0
28-31.9	61	5.5	26.5	914	4.6
32-35.9	54	3.8	23.5	873	4.4
36-39.9	42	2.4	18.3	724	3.6
40-43.9	30	1.4	13.0	566	2.8
44-47.9	18	0.7	7.8	344	1.7
48-51.9	18	0.6	7.8	345	1.7
52-55.9	14	0.4	6.1	311	1.6
56-59.9	8	0.2	3.5	153	0.8
60-63.9	14	0.3	6.1	321	1.6
All	522	72.5	227.0	7915	39.7

TABLE 3. STOCK TABLE BY SPECIES, PER ACRE

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	164	22.5	71.3	24.1	2504	12.6
PP	149	20.1	64.8	24.3	2487	12.5
IC	85	13.1	37.0	22.8	1064	5.3
DF	76	10.2	33.0	24.4	1079	5.4
SP	48	6.5	20.9	24.2	780	3.9
All	522	72.5	227.0	24.0	7915	39.7

Trees: tally trees. TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species codes: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

A map of the study area, which is an irregularly shaped lake or reservoir. The area is divided into 48 numbered sampling points, represented by blue dots with black outlines. The points are distributed across the lake, with some points clustered together and others more isolated. The numbering starts at 01 in the lower-left and proceeds generally towards the upper-right, with some points like 09, 11, 18, and 25 located near the top edge. The lake's boundary is defined by a thick black line.

QA findings: none; unit accepted

TABLE 4. PLOT LIST

Plot	Trees in	BA sq ft/ac	Slope %	Status
107-01	13	260	11	ok
107-02	12	240	5	ok
107-03	13	260	31	ok
107-04	11	220	12	ok
107-05	8	160	20	ok
107-06	11	220	6	ok
107-07	11	220	9	ok
107-08	13	260	6	ok
107-09	12	240	21	ok
107-10	10	200	17	ok
107-11	12	240	18	ok
107-12	11	220	16	ok
107-13	10	200	20	ok
107-14	13	260	39	ok
107-15	13	260	19	ok
107-16	11	220	15	ok
107-17	9	180	23	ok
107-18	12	240	9	ok
107-19	12	240	25	ok
107-20	10	200	29	ok
107-21	12	240	22	ok
107-22	12	240	23	ok
107-23	11	220	29	ok
107-24	12	240	9	ok
107-25	10	200	27	ok
107-26	8	160	28	ok
107-27	10	200	15	ok
107-28	11	220	26	ok
107-29	12	240	36	ok
107-30	11	220	8	ok
107-31	11	220	16	ok
107-32	12	240	25	ok
107-33	10	200	29	ok
107-34	11	220	12	ok
107-35	11	220	17	ok
107-36	11	220	14	ok
107-37	12	240	22	ok
107-38	12	240	34	ok
107-39	12	240	30	ok
107-40	12	240	20	ok
107-41	12	240	13	ok
107-42	13	260	30	ok
107-43	14	280	33	ok
107-44	10	200	23	ok
107-45	11	220	18	ok
107-46	12	240	16	ok